Math Thermo
class # 03
01/27/2026

Differential Notation

In the physics literature it is common to write

(3.1) $$d\tilde{U}^\alpha = T_v^\alpha dS^\alpha - P_v^\alpha dV^\alpha + \sum_{i=1}^{r} \mu_{v,i}^\alpha dN_i^\alpha$$

and

(3.2) $$d\tilde{S}^\alpha = \frac{1}{T_s^\alpha} dU^\alpha + \frac{P_s^\alpha}{T_s^\alpha} dV^\alpha - \sum_{i=1}^{r} \frac{\mu_{s,i}^\alpha}{T_s^\alpha} dN_i^\alpha .$$

(3.1) means $\tilde{U}^\alpha = \tilde{U}^\alpha(S, V, \vec{N})$ and

$$\frac{\partial \tilde{U}^\alpha}{\partial S^\alpha} = T_v^\alpha, \quad \frac{\partial \tilde{U}^\alpha}{\partial V^\alpha} = -P_v^\alpha, \quad \frac{\partial \tilde{U}^\alpha}{\partial N_j^\alpha} = \mu_{v,i}^\alpha$$

and (3.2) simply means $\tilde{S}^\alpha = \tilde{S}^\alpha(U^\alpha, V^\alpha, \vec{N}^\alpha)$ with

$$\frac{\partial \tilde{S}^\alpha}{\partial U^\alpha} = \frac{1}{T_s^\alpha}, \quad \frac{\partial \tilde{S}^\alpha}{\partial V^\alpha} = \frac{P_s^\alpha}{T_s^\alpha}, \quad \frac{\partial \tilde{S}}{\partial N_j^\alpha} = -\frac{\mu_{s,i}^\alpha}{T_s^\alpha}$$

We will ascribe real meaning to these relations, once we define process paths, using the chain rule.

Path (Contour) Integrals

Defn (3.1): Suppose that $D \subseteq \mathbb{R}^n$ is open. A function $\vec{\gamma}: [a, b] \to D$ is called a path (or contour) iff $\vec{\gamma}$ is continuous and piecewise smooth. D is called path-connected iff for every two distinct points $\vec{a}, \vec{b} \in D$ there is path $\vec{\gamma}: [a, b] \to D$ such that

$$\vec{\gamma}(a) = \vec{a} \qquad \vec{\gamma}(b) = \vec{b}.$$

D is called <u>simply-connected</u> iff it is (1) path connected and (2) path can be continuously deformed to a point, i.e., there are no holes. D is called <u>convex</u> iff for every pair $\vec{a}, \vec{b} \in D$, the point

$$\vec{x}(t) = \vec{a}(1-t) + \vec{b}t \in D$$

for all $t \in [0,1]$.

<u>Definition</u> (3.2): Let $\vec{F}: D \to \mathbb{R}^n$ be a C^1 function, i.e., $\vec{F} \in C^1(D;\mathbb{R}^n)$. Let $\vec{\gamma}:[a,b] \to D$ path in D, which is assumed to be simply connected. Then the <u>path integral</u> $\int_{\vec{\gamma}} \vec{F}(\vec{x}) \cdot d\vec{x}$ is defined via

(3.3) $$\int_{\vec{\gamma}} \vec{F}(\vec{x}) \cdot d\vec{x} := \int_a^b \vec{F}(\vec{\gamma}(\tau)) \cdot \vec{\gamma}'(\tau)\, d\tau.$$

We will also use the notation

$$\int_{\vec{\gamma}} \vec{F}(\vec{x}) \cdot d\vec{x} = \int_{\vec{\gamma}} F_1(\vec{x})dx_1 + \cdots + F_n(\vec{x})dx_n$$

<u>Defn</u> (3.3): Let $D \subseteq \mathbb{R}^n$ be a simply connected open set. A path $\vec{\gamma}:[a,b] \to D$ is called <u>closed</u> iff

$$\vec{\gamma}(a) = \vec{\gamma}(b)$$

A closed path is called <u>simple</u> iff it does not intersect itself except at $\tau = a$ and $\tau = b$, i.e., for every $c \in (a,b)$

$$\vec{\gamma}(c) \neq \vec{\gamma}(\tau), \qquad \tau \in [a,c) \cup (c,b]$$

<u>Theorem</u> (3.4): Let $D \subseteq \mathbb{R}^n$ be an open, simply-connected set. Assume $\vec{\gamma}:[a,b] \to D$ is a path. If $\vec{x}:[c,d] \to D$ is a path in D, with the property that

$$\vec{x}(c) = \vec{\gamma}(a), \qquad \vec{x}(d) = \vec{\gamma}(b),$$

and

$$\text{Range}(\vec{x}) = \text{Range}(\vec{\gamma}),$$

then

$$\int_{\vec{\gamma}} \vec{F}(\vec{x}) \cdot d\vec{x} = \int_{\vec{x}} \vec{F}(\vec{x}) \cdot d\vec{x}.$$

This result guarantees that the path integrals are parametrization independent.

If $C = \text{Range}(\vec{\gamma}) = \text{Range}(\vec{x})$, then we write

$$\int_{C} \vec{F}(\vec{x}) \cdot d\vec{x} = \int_{\vec{\gamma}} \vec{F}(\vec{x}) \cdot d\vec{x}.$$

<u>Dfn</u> (3.5): Let $D \subseteq \mathbb{R}^n$ be an open, simply connected set. Suppose that

$$\int_{\vec{\gamma}_1} \vec{F}(\vec{x}) \cdot d\vec{x} = \int_{\vec{\gamma}_2} \vec{F}(\vec{x}) \cdot d\vec{x}$$

for any two paths $\vec{\gamma}_1 : [a,b] \to D$, $\gamma_2 : [c,d] \to D$ with

$$\vec{\gamma}_1(a) = \vec{\gamma}_2(a) \quad \text{and} \quad \vec{\gamma}_1(b) = \vec{\gamma}_2(b).$$

Then we say that the integral is <u>path independent</u>. Note that we are <u>not</u> assuming that

$$\text{Range}(\vec{\gamma}_1) = \text{Range}(\vec{\gamma}_2).$$

<u>Defn</u> (3.6): Let D be an open set and $\vec{F} \in C^1(D;\mathbb{R}^n)$. We say that $\vec{F}$ is <u>conservative</u> iff there is a function $f \in C^2(D;\mathbb{R})$ such that

$$\vec{F}(\vec{x}) = \nabla f(x), \quad \forall\ \vec{x} \in D.$$

<u>Theorem</u> (3.7): Let D be an open simply connected set in $\mathbb{R}^n$. If $\vec{F} \in C^1(D,\mathbb{R}^n)$ is conservative, then the integral

$$\int_{\vec{\gamma}} \vec{F}(\vec{x}) \cdot d\vec{x}$$

is path-independent.

<u>Proof</u>: Let $\vec{\gamma}_1 : [a_1, b_1] \to D$ and $\vec{\gamma}_2 : [a_2, b_2] \to D$ be paths in D with the same end points, i.e.,

$$\vec{a} := \vec{\gamma}_1(a_1) = \vec{\gamma}_2(a_2), \quad \vec{\gamma}_1(b_1) = \vec{\gamma}_2(b_2) =: \vec{b}$$

By the chain rule, for $i = 1, 2$,

$$\frac{d}{d\tau} f(\vec{\gamma}_i(\tau)) = \nabla f(\vec{\gamma}_i(\tau)) \cdot \vec{\gamma}_i'(\tau)$$

$$= \vec{F}(\vec{\gamma}_i(\tau)) \cdot \vec{\gamma}_i'(\tau)$$

Thus,

$$\int_{\vec{\gamma}_i} \vec{F}(\vec{x}) \cdot d\vec{x} = \int_{a_i}^{b_i} \vec{F}(\vec{\gamma}_i(\tau)) \cdot \vec{\gamma}_i'(\tau)\, dt$$

$$= \int_{a_i}^{b_i} \frac{d}{d\tau}\left[f(\vec{\gamma}_i(\tau))\right] d\tau$$

$$\overset{FTC}{=} f(\vec{b}) - f(\vec{a}),$$

for $i = 1, 2$.

We have the following well-known results.

<u>Theorem</u> (3.8): Let $D \subseteq \mathbb{R}^n$ be a simply-connected set and suppose that $\vec{F} \in C^1(D; \mathbb{R}^n)$. The following are equivalent

1) $\vec{F}$ is conservative

2) $\int_{\vec{\gamma}} \vec{F}(\vec{x})\, d\vec{x}$ is path independent

3) $\oint_{\vec{\gamma}} \vec{F}(\vec{x})\, d\vec{x} = 0$ for any closed path.

Recall, we wrote, as a short hand

$$d\tilde{U} = T_U\, dS - P_U\, dV + \sum_{i=1}^{r} \mu_{U,i}\, dN_i$$

This has the form

$$F_1\, dx_1 + F_2\, dx_2 + \cdots + F_n\, dx_n = \vec{F} \cdot d\vec{x}$$

where

$$F_1 = T_U, \quad F_2 = -P_U, \quad \ldots$$

We say that $\vec{F} \cdot d\vec{x}$ is an <u>exact differential</u> if $\vec{F}$ is conservative.

Clearly

$$d\tilde{U} = \nabla\tilde{U} \cdot d\vec{\sigma} \qquad (\vec{\sigma} \in \Sigma_s)$$

is an exact differential, because $\nabla\tilde{U}$ is conservative, trivially. Thus, the integral

$$\int_{\vec{\gamma}} d\tilde{U} = \int_{\vec{\gamma}} \nabla\tilde{U} \cdot d\vec{\sigma}$$

is path independent. If $\vec{\gamma} : [a,b] \to \Sigma_s$ is the path in question, with

$$\vec{\gamma}(a) = \vec{\sigma}_a, \qquad \vec{\gamma}(b) = \vec{\sigma}_b.$$

then

$$\int_{\vec{\gamma}} d\tilde{U} = \int_{\vec{\gamma}} \nabla\tilde{U} \cdot d\vec{\sigma} = \tilde{U}(\vec{\sigma}_b) - \tilde{U}(\vec{\sigma}_a).$$

We can use any path we want in state space Σ_s

The same is true for

$$d\tilde{S} = \frac{1}{T_s} dU + \frac{P_s}{T_s} dV - \sum_{i=1}^{r} \frac{\mu_{s,i}}{T_s} dN_i,$$

that is

$$d\tilde{S} = \nabla\tilde{S} \cdot d\vec{\sigma} \qquad (\vec{\sigma} \in \Sigma_v)$$

is an exact differential.

Heat Flow

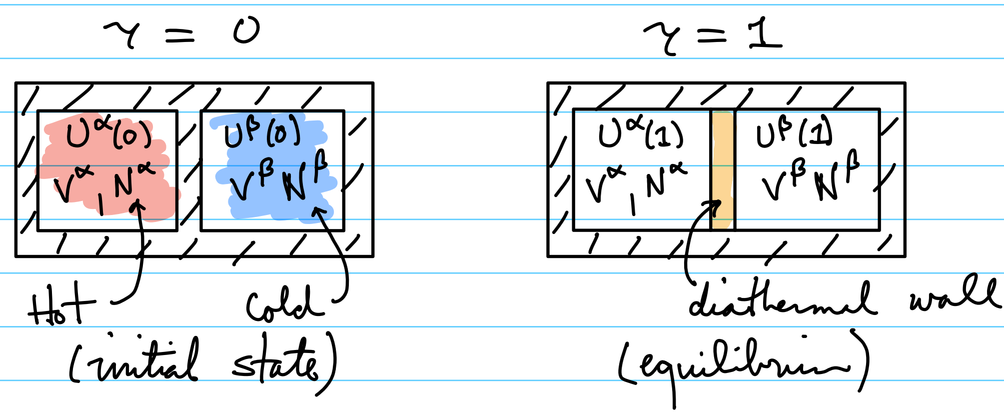
Let us examine the approach to thermal equilibrium using a diathermal wall. Suppose the substance is unary, $r=1$. It must be that the variables

$$V^\alpha, V^\beta, N^\alpha, N^\beta$$

are fixed. But energy can be exchanged in the process.

Suppose the variable τ parameterizes the process.

At $\tau = 0$, we replace the insulating wall with a diathermal wall



$$S(0) = \overbrace{\tilde{S}^\alpha(U^\alpha(0), V^\alpha, N^\alpha) + \tilde{S}^\beta(U^\beta(0), V^\beta, N^\beta)}^{\text{initial state}}$$

$$< \underbrace{\tilde{S}^\alpha(U^\alpha(1), V^\alpha, N^\alpha) + \tilde{S}^\beta(U^\beta(1), V^\beta, N^\alpha)}_{\text{equilibrium state}} = S(1)$$

We require that

$$U^\alpha(\tau) + U^\beta(\tau) = U_0$$

The entropy is at a max at equilibrium.

Suppose that, as indicated by the figure above,

$$T^\alpha(U^\alpha(0), V^\alpha, N^\alpha) > T^\beta(U^\beta(0), V^\beta, N^\beta)$$

we know that when equilibrium is attained

$$T^\alpha(U^\alpha(1), V^\alpha, N^\alpha) = T^\beta(U^\beta(1), V^\beta, N^\beta)$$

We will show that

$$U^\alpha(0) > U^\alpha(1) \quad \text{(energy is lost in } \alpha)$$

and

$$U^\beta(0) < U^\beta(1) \quad \text{(energy is gained in } \beta)$$

To see this compute

$$\frac{dS}{d\tau} = \frac{\partial \tilde{S}^\alpha}{\partial U^\alpha}\frac{\partial U^\alpha}{\partial \tau} + \frac{\partial \tilde{S}^\beta}{\partial U^\beta}\frac{\partial U^\beta}{\partial \tau}$$

$$= \frac{1}{T_s^\alpha}\frac{\partial U^\alpha}{\partial \tau} + \frac{1}{T_s^\beta}\frac{\partial U^\beta}{\partial \tau}$$

(3.4) $$= \left(\frac{1}{\tilde{T}_s^\alpha(\tau)} - \frac{1}{\tilde{T}_s^\beta(\tau)}\right)\frac{\partial U^\alpha}{\partial \tau}$$

where

$$\tilde{T}_s^q(\tau) := T_s^q(U^q(\tau)), \qquad q = \alpha, \beta.$$

We know that

$$S(0) \leq S(1).$$

We know that

$$\tilde{T}_s^{\alpha}(0) > \tilde{T}_s^{\beta}(0)$$

and

$$\tilde{T}_s^{\alpha}(1) = T_s^{\beta}(0).$$

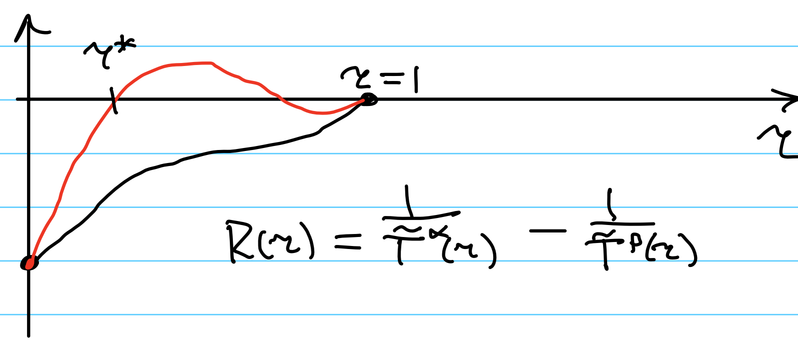
Set

$$R(\tau) := \frac{1}{\tilde{T}^{\alpha}(\tau)} - \frac{1}{\tilde{T}^{\beta}(\tau)} \qquad 0 \leq \tau \leq 1$$

We know that

$$R(0) < 0 \quad \text{and} \quad R(1) = 0$$

Thus



The red curve is not possible; if it were equilibrium would be reach earlier, at $\tau = \tau^*$.
Thus

$$R(\tau) < 0 \quad \forall \tau \in [0, 1).$$

Integrating equation (3.4), we get

(3.5) $$0 \leq S(1) - S(0) = \int_0^1 R(\tau) \frac{dU^{\alpha}}{d\tau} d\tau$$

For $U^\alpha(\tau)$ we have two options

$$\text{Case}(1) \quad U^\alpha(0) > U^\alpha(1)$$

or

$$\text{Case}(2) \quad U^\alpha(0) \leq U^\alpha(1).$$

We are free to pick a parameterization however we want. Let us take a simple linear path

$$\frac{dU^\alpha}{d\tau} = \frac{U^\alpha(1) - U^\alpha(0)}{1} =: C^\alpha$$

Then

$$\text{Case}(1) \implies C^\alpha < 0$$

$$\text{Case}(2) \implies C^\alpha \geq 0$$

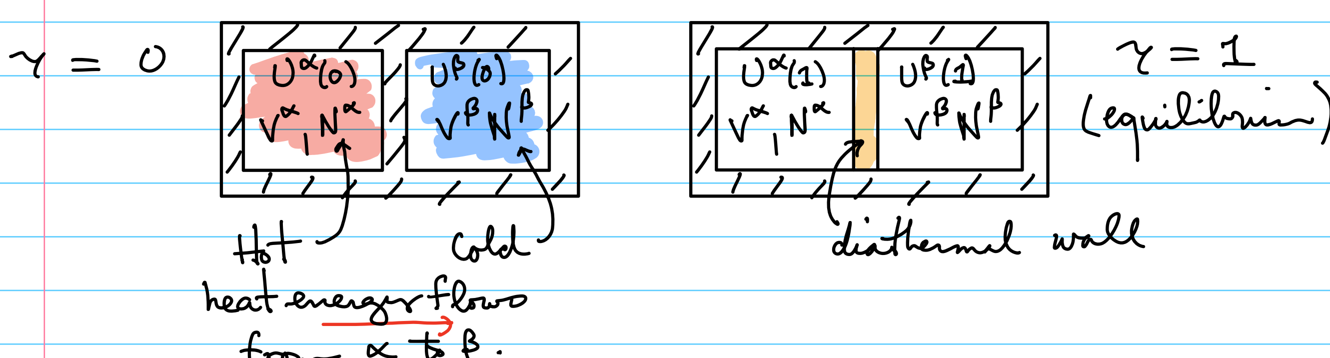
The integral in (3.5) is

$$0 \leq S(1) - S(2) = C^\alpha \underbrace{\int_0^1 R(\tau)\, d\tau}_{<0}.$$

Therefore, the only possible choice is Case (1). Thus

$$\boxed{U^\alpha(0) > U^\alpha(1) \qquad \left(U^\beta(0) < U^\beta(1)\right).}$$

This is consistent with our intuition about temperature. If $T^\alpha(0) > T^\beta(0)$ heat energy flows from subsystem α to subsystem β:



Heat Transfer Principle: Heat energy always flows from hotter to colder systems. This is equivalent to the second Law of Thermodynamics.